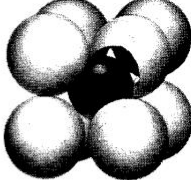
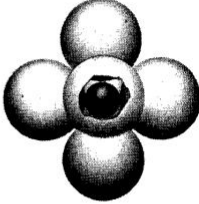
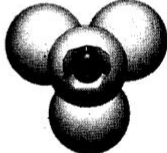
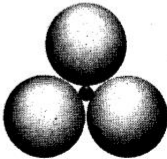


MSE 317: Ceramic Materials

Homework 1 Due: Sep 20, 2022

Student Name _____ Student No. _____ Points _____

1. Verify the limiting radius ratios for 3, 4, 6, and 8-fold coordinations that are given in the following figure. (20 points)

Coordination Number	Disposition of Ions about Central Ion	Range of – Cation Radius Ratio Anion Radius	
8	Corners of cube	≥ 0.732	
6	Corners of octahedron	≥ 0.414	
4	Corners of tetrahedron	≥ 0.225	
3	Corners of triangle	≥ 0.155	

2. Calculate the lattice constants and density of fluorite. $r(\text{Ca}^{2+})=100$ pm, $r(\text{F}^-)=133$ pm.(20 points)